

MATH 535: LINEAR ALGEBRA

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CONTENTS

1. August 24, 2026	1
1.1. Fields	1

1. AUGUST 24, 2026

1.1. Fields.

Definition 1 (Binary operation). *A binary operation on a set S is a function*

$$F: S \times S \longrightarrow S.$$

Thus, for every $x, y \in S$, the value $F(x, y)$ is again an element of S . We usually write $x \cdot y$ or $x + y$ in place of $F(x, y)$.

Example 1. *Addition on $\mathbb{Z}$ is a binary operation: the map*

$$\mathbb{Z} \times \mathbb{Z} \longrightarrow \mathbb{Z}, \quad (m, n) \longmapsto m + n$$

takes every pair of integers to an integer.

Definition 2. *Let F be a binary operation on S . A subset $T \subseteq S$ is closed under F if*

$$F(x, y) \in T$$

whenever $x, y \in T$.

Example 2. *The positive integers $\mathbb{Z}_{>0}$ are closed under addition. The odd integers are not closed under addition, but they are closed under multiplication.*

Definition 3 (Field). Let $\mathbb{F}$ be a set equipped with two binary operations, called addition and multiplication. The sum and product of $a, b \in \mathbb{F}$ are denoted by $a + b$ and ab , respectively. Suppose that for every $a, b, c \in \mathbb{F}$,

- (i) $a + b = b + a$;
- (ii) $(a + b) + c = a + (b + c)$;
- (iii) $ab = ba$;
- (iv) $a(bc) = (ab)c$;
- (v) $a(b + c) = ab + ac$.

Suppose further that

- (vi) $\mathbb{F}$ contains an additive identity 0 and a multiplicative identity 1 , so that

$$a + 0 = a \quad \text{and} \quad 1a = a$$

for every $a \in \mathbb{F}$;

- (vii) $1 \neq 0$;
- (viii) every $a \in \mathbb{F}$ has an additive inverse $-a \in \mathbb{F}$ such that $a + (-a) = 0$;
- (ix) every nonzero $a \in \mathbb{F}$ has a multiplicative inverse $a^{-1} \in \mathbb{F}$ such that $aa^{-1} = 1$.

Then $\mathbb{F}$ is called a field.

Example 3. The integers $\mathbb{Z}$ do not form a field: for example, 2 has no multiplicative inverse in $\mathbb{Z}$. The sets $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$, with their usual operations, are fields.

There are also finite fields. For example,

$$\mathbb{F}_2 = \{0, 1\} \quad \text{and} \quad \mathbb{F}_3 = \{0, 1, 2\},$$

where the elements are the possible remainders upon division by 2 and 3, respectively. Addition and multiplication in $\mathbb{F}_3$ are given by the

following tables:

$+$	0	1	2	$\cdot$	0	1	2
0	0	1	2	0	0	0	0
1	1	2	0	1	0	1	2
2	2	0	1	2	0	2	1

More generally, if p is prime, then

$$\mathbb{F}_p = \{0, 1, \dots, p-1\},$$

with addition and multiplication performed modulo p , is a field.

Lemma 1. *If a is an element of a field $\mathbb{F}$, then $a \cdot 0 = 0$.*

Proof. Since $0 + 0 = 0$, distributivity gives

$$a \cdot 0 = a(0 + 0) = a \cdot 0 + a \cdot 0.$$

Adding the additive inverse of $a \cdot 0$ to both sides yields $0 = a \cdot 0$. $\square$